Decentralizing Agriculture

A shift in the industrial paradigm of food production

Marco Zolla Salazar

AuthorId

Faculty

Major

bei

Employer

Kontakt:

marco.zolla@proton.me

ContactTitle

Contents

Abstract

Industrial agriculture is responsible for the majority of global biodiversity loss, roughly one third of all greenhouse gas emissions, and the creation of fragile, highly concentrated food supply chains on which billions of people existentially depend. This paper proposes an open-source controlled environment agriculture (CEA) platform designed to address these problems through decentralisation, automation, and a shared data commons.

The system comprises a sensor array and camera mounted in a small-scale greenhouse, a microcontroller running a PID control loop over environmental actuators, a single-board computer performing edge ML inference, and an optional connection to a cloud-based data flywheel where growth models are continuously retrained on anonymised telemetry from the global network of deployed nodes. A computer vision pipeline provides growth metrics, disease diagnosis, and species recognition. A climate-aware species recommendation engine minimises energy consumption by matching recommended crops to the local ambient environment. All hardware designs, firmware, software, trained models, and collected data are released under open licences, enabling a virtuous cycle: the more nodes deployed, the richer the shared dataset, and the more accurate the growth optimisation.

The project does not aim to replace field agriculture wholesale. Its niche is high-value, perishable, water-intensive crops grown in proximity to consumers — the segment where the environmental and supply-chain benefits of CEA are largest and most defensible. Within that niche, we argue the platform can meaningfully reduce land use pressure, water consumption, fertiliser runoff, pesticide pollution, transport emissions, food waste, and supply-chain concentration, while enabling a diversity of crops that the industrial monoculture model structurally cannot support.

1 Introduction

1.1 Motivation

The global food system is in a state of structural crisis. Industrial agricultural production — the dominant paradigm for the past century — has delivered extraordinary increases in caloric output, but at costs that are becoming impossible to ignore: ecosystem collapse, soil exhaustion, water depletion, climate destabilisation, and the creation of food supply chains so concentrated that a single geopolitical shock can threaten the nutrition of hundreds of millions of people.

At the same time, the tools to build something better have never been more accessible. Capable microcontrollers cost a few dollars. Machine learning frameworks are open source. Camera modules capable of running computer vision inference are commodity hardware. The question is no longer whether small-scale, intelligent, automated food production is technically feasible — it is — but whether it can be made cheap enough, robust enough, and open enough to displace the incumbent system at meaningful scale.

This paper describes a platform designed to make that possible.

1.2 Project vision

The project's core proposition is that a network of many small, locally operated, data-connected greenhouses can collectively outperform a system of few large, centralised farms — on environmental metrics, on resilience, and eventually on cost — provided that the intelligence required to operate them efficiently is shared as a common good rather than locked up in proprietary systems.

The platform is designed around four interlocking principles:

1. **Edge-first control.** All critical control logic runs locally and offline. The system operates without internet connectivity. The cloud connection is optional and asynchronous, used only for learning and coordination, never for real-time operation.

2. **Open everything.** Hardware reference designs, firmware, software, trained models, and the collected dataset are all released under maximally permissive open licences. Anyone can deploy, modify, and contribute back.

3. **Data flywheel.** Every deployed node, with the user's consent, contributes anonymised telemetry to a shared dataset. Models trained on this dataset are continuously improved and pushed back to all nodes via over-the-air updates. The network becomes smarter as it grows.

4. **Climate-adapted intelligence.** The system does not prescribe a fixed set of crops. A climate-aware recommendation engine suggests species whose optimal growing conditions most closely match the local ambient environment, minimising the energy required to close the gap between what the environment provides and what each crop needs.

2 The problems of industrial agriculture

Although the rise of the modern industrial structure of agricultural production has given many societies the ability to forestall famine and thrive in ways long thought impossible, many of its social and ecological repercussions have revealed themselves to be potentially dangerous in the best scenario and significant contributors to an existential crisis in the worst. Every step throughout the production process results in ecological problems of varying severity, from the clearing of the land to the delivery of the final product, each impacting the surrounding ecosystems and underlying soils in different ways. It is a production model that forces the whole population into the dependence of a few producers, who have increasingly incorporated more and more monoculture practices, thus making the food supply more vulnerable to collapse in the case of an uncontrolled pest. The following sections document each of these problems in turn, drawing on recent peer-reviewed literature and institutional reports.

2.1 Land Use

The prevailing paradigm of industrial agricultural production requires the use of vast swathes of land, along with a considerable amount of water, energy and other resources, which inevitably ends up homogenizing the local flora and destroying the habitat of the local fauna, leading to an overall loss of biodiversity. The agricultural expansion that the current system entails is indeed one of the main drivers of global biodiversity loss [1, p. 412], [2, p. 131], and is the most widespread form of land-use change, with cropping and animal husbandry covering over one third of the terrestrial land surface [1, p. XVI]. More than 90% of global biodiversity impacts from land-use change are attributable to agriculture, with crop cultivation accounting for 72% and pastures for 21% — dwarfing the combined impact of mining, urban development, and all other industries [3].

This expansion of cropland has been a constant for decades, with global cropland area growing by 78 million ha (780,000 km^2) between 2001 and 2023, largely driven by an increase in permanent crops such as cocoa, oil palm and coffee [4, p. 3]. All of these are used intensively for monoculture, as are feedstock crops—as of the early 2000s, only 55% of global crop calories were allocated to direct human consumption, with 36% diverted to animal feed and the remainder to biofuels and industrial uses [5, Table 2]; this particular form of agriculture, which consists in the use of a single crop for large productions, is another potent driver of biodiversity and habitat loss which will be explored in a subsequent section.

The extension of anthropogenic drivers of change has led to as little as 13% of the ocean and 23% of the land falling under the classification of 'wilderness'. These few remaining areas, classified as sufficiently free of obvious human impacts to qualify as wilderness, tend to be remote or unproductive, like tundras or oceanic gyres, the most hospitable biomes having been almost or completely modified by humans in most regions [1, p. 206]. With the transformation for human convenience of the landscapes of the world that are the most accommodating to life, displaced wild populations are left without a habitat.

The impact of this intrusion into the wilderness has far-reaching consequences as well. According to the FAO's State of Food and Agriculture report [6, p. 4], agriculture drives nearly 90 percent of global deforestation, with cropland expansion accounting for almost half of the total deforested area. This leads to changes in both global and local climates through two main mechanisms: the release of stored carbon produces a robust warming signal worldwide, while the removal of forest cover changes surface albedo, evapotranspiration and canopy roughness, which produces a net global warming effect for deforestation in the tropics and mid-latitudes, but a net cooling effect beyond roughly 50°N where increased albedo dominates. At the local scale, deforestation in the tropics raises surface temperatures by 0.2–2.4°C, with smaller increases in temperate zones (0.02–1.0°C), though boreal deforestation can produce modest local cooling [7, pp. 1, 6].

A pan-tropical synthesis estimated total tropical deforestation at 6.5–9.5 million hectares per year between 2011 and 2015, of which agriculture drove the overwhelming majority (90–99%), accounting for 6.4–8.8 million hectares per year [8, pp. 1, 3]. Crucially, only 45–65% of all deforested land was subsequently converted into actively managed cropland, pasture, or tree crops, meaning that a substantial fraction of tropical forest destruction does not result in lasting agricultural production.

2.2 Biodiversity loss

Beyond direct habitat destruction, the intensification of agriculture has driven rapid declines among species of agricultural habitats, contributing to sharply rising extinction rates across all taxonomic groups for which robust assessment data exist [1, Section 2.2.5.1]. The spread and intensification of agriculture over the past half century — through greatly expanded scales, monoculturing, increasing pesticide and fertiliser application, and the elimination of hedgerows and other wildlife habitat — is directly related to major documented declines in insect biomass and diversity, with climate change—to which agriculture is itself a leading contributor (Section 2.8)—acting as an additional stressor [9, pp. 1–3]. At the global scale, agriculture has been identified as a threat to 24,000 of the 28,000 (86%) species so far documented as at risk of extinction [10, p. 7].[1]

The impact reaches beyond field boundaries. A 27-year study using standardised traps across 63 German nature protection areas found a seasonal decline of 76% and a mid-summer peak decline of 82% in total flying insect biomass, with losses apparent regardless of habitat type, suggesting large-scale drivers consistent with, though not yet definitively attributable to, agricultural intensification [11, pp. 14–15]. This decline is multi-causal and compounding: habitat loss, climate change, pesticide exposure, and other stressors co-occur and interact, with reported rates of annual abundance decline frequently around 1–2% across taxa, though with much variation across time, space, and taxonomic lineage [12, p. 2]. A global analysis of 923 terrestrial insect assemblages monitored across 106 long-term studies found overall abundance declining at approximately 1.5% per year, with losses driven disproportionately by formerly abundant species, though data remain heavily concentrated in Europe and North America [13, Fig. 2]. Spatially, the interaction of intensive agriculture and climate warming is associated with reductions of almost 50% in abundance and 27% in species richness relative to undisturbed habitats [14, Fig. 2]. In the tropics, where long-term monitoring data are scarcer, a 22-year study in protected Costa Rican lowland forest documented declines in 41% of common caterpillar genera, with losses linked to climate variability and, potentially, pesticide drift from adjacent agricultural land [15, pp. 3, 6].

[1]The Chatham House figure draws on IUCN Red List data as compiled by Ritchie and Roser (2019).

The decline extends well beyond insects. A 37-year monitoring programme spanning 170 common bird species across more than 20,000 sites in 28 European countries found an overall population decline of 25%, with farmland birds suffering the steepest losses at nearly 57%—far exceeding declines in woodland, urban, or other habitat-associated species. Agricultural intensification, measured as the cover of farms with high pesticide and fertiliser inputs, was identified as the dominant pressure in a quasicausal analysis [16, Fig. 2]. Pollinators have been similarly affected: managed honey bee colonies declined by 59% in North America between 1947 and 2005 and by 25% in central Europe between 1985 and 2005, though global managed stocks rose by approximately 45% between 1961 and 2008 due to expansion in countries such as China and Argentina; among wild bees, four bumblebee species have gone extinct across Europe [17, p. 1255957-1]. The mechanistic drivers of pollinator loss—including neonicotinoid exposure, habitat destruction, and pathogen interactions—are examined in Section 2.5.

2.3 Soil degradation and erosion

Soils underpin the production of ecosystem goods and services that sustain both natural habitats and human societies. They are a key reservoir of biodiversity, from micro-organisms to flora and fauna; and this biotic community drives the soil functions (nutrient cycling, water regulation, carbon sequestration, etc.) that generate those services. Those functions are controlled primarily by the soil's chemical composition, biological activity, and physical structure. When any of these attributes deteriorate, the soil's ability to deliver essential ecosystem services, such as food production, water purification, and climate regulation, is compromised or lost [18, pp. 4–5].

Industrial tillage-based agriculture destroys soil at a rate that outpaces natural soil formation by orders of magnitude. Erosion rates from conventionally ploughed agricultural fields average one to two orders of magnitude greater than rates of soil production, erosion under native vegetation, and long-term geological erosion — making conventional plow-based agriculture fundamentally unsustainable over civilizational timescales [19]. Industrial farming practices such as mono-cropping, minimal use of organic fertilisers[2], and removal of crop residues are a primary cause of the decline in soil organic matter, which is essential to long-term soil productivity [20, Table 7].

The IPCC's Special Report on Climate Change and Land estimates that cropland soils have lost 20–60% of their organic carbon content relative to pre-cultivation states (medium confidence), and that soils under conventional agriculture continue to be a source of greenhouse gas emissions (medium confidence) [21, TS.4].

A second-order meta-analysis synthesizing 230 first-order meta-analyses encompassing more than 25,000 primary studies confirms that land conversion for crop production is a consistently net-negative driver for soil organic carbon (SOC), with mean SOC losses of approximately 25% following conversion from forest or wetland, and 16% from grassland (though the latter carries a wider confidence interval that includes zero), though forest-conversion losses in particular may be underestimated because most underlying studies measure only over years to decades, whereas full equilibration after land-use change may take considerably

[2]Organic fertilisers — manure, compost, and crop residues returned to the field — add carbon-rich matter that sustains soil biological activity and structure. They are distinct from the synthetic nitrogen and phosphorus fertilisers whose environmental impacts are discussed in Section 2.4.

longer (estimated at around 80 years for grassland-to-cropland conversion). This matters because SOC is foundational to soil health, food production, and climate regulation: global SOC stocks to 2 m depth are estimated at approximately 2,400 Gt C — three times the carbon currently held in the atmosphere — meaning even small proportional losses translate into significant atmospheric CO_2 increases. SOC restoration is possible, but it is understood to be generally slower than depletion [22].

The on-site consequences of this depletion are direct: SOC loss diminishes nutrient availability, water-holding capacity, and soil structural stability, reducing the productive capacity of cropland and driving increased dependence on synthetic inputs to maintain yields [20, pp. 4–5]. The agronomic mechanisms by which monoculture accelerates this degradation, and the yield recovery achievable through diversification, are examined in Section 2.6.3.

2.4 Fertilizer use

Excess nitrogen and phosphorus from agricultural fertilizers leach into waterways, triggering eutrophication — a process in which accelerated algal growth reduces water clarity and, when these blooms die, microbial decomposition severely depletes dissolved oxygen, creating hypoxic or anoxic dead zones in which most aerobic organisms cannot survive [23, Consequences]. Cultural eutrophication of this kind has been identified as a leading cause of freshwater ecosystem impairment [24, p. 1] and has driven the exponential spread of coastal dead zones, from over 400 documented systems in 2008 [25, p. 926] to more than 500 by 2018 [26, p. 1].

Between 50 and 70% of applied nitrogen is typically lost to the environment through volatilisation, leaching, surface runoff, and denitrification [27, p. 1]. These pathways drive the eutrophication described above — through nutrient-laden runoff and leachate reaching waterways — and the atmospheric accumulation of nitrous oxide examined below, as excess soil nitrogen undergoes microbial denitrification.

Excess nitrogen in agricultural soils undergoes microbial conversion to nitrous oxide (N_2O) — a greenhouse gas with a global warming potential 273 times that of CO_2 over 100 years [28, Table 7.15]. Global anthropogenic N_2O emissions increased by 40% between 1980 and 2020, driven primarily by agricultural nitrogen use [29, Section 4.2.1], and atmospheric concentrations have now exceeded projected levels under all CMIP5 and CMIP6 scenarios [29, Section 4.1.1].

Critically, the relationship between nitrogen application and N_2O emissions is not linear but accelerating: as fertilizer inputs increase beyond crop demand, the emission factor itself grows, meaning that each additional unit of nitrogen applied in excess generates a disproportionately larger N_2O penalty. As a consequence, small reductions in over-application can yield emissions reductions substantially larger than linear models — such as the IPCC's default 1% emission factor — would predict [30, pp. 9199–9202]. Notably, this nonlinear amplification is most pronounced for conventional synthetic fertilizers; controlled-release formulations show a significantly attenuated response [30, p. 9201] — consistent with slower nitrogen delivery reducing the excess that drives nonlinear emissions.[3] In soilless systems such as closed-loop hydroponics,

[3] The evidence base for controlled-release attenuation of nonlinear N_2O response remains small ($n = 8$ site-years in the meta-analysis). Instrumented soil-based units within the network proposed in this paper could contribute longitudinal field data to strengthen or qualify this finding.

the microbial nitrification–denitrification pathway that generates these emissions is largely absent, removing the mechanism entirely rather than merely attenuating it.

2.5 Pesticide use

Pesticides contaminate soil, water, and air well beyond the target application zone, harming non-target organisms throughout the food web. An estimated 64% of the world's agricultural land — covering approximately 24.5 million km^2 — is at risk of pollution by more than one pesticide active ingredient, with 31% of that area at high risk; five watersheds identified as presenting high combined concern for pesticide pollution risk, water scarcity, and biodiversity include the Orange (South Africa), Huang He (China), Indus (India), Murray (Australia), and Paraná (Argentina) [31, pp. 206–208].

A 2023 meta-analysis synthesizing 54 studies found that pesticides consistently reduced both the abundance and diversity of soil fauna communities. The most damaging scenarios involved combinations of multiple substances, broad-spectrum compounds, and insecticides, which significantly decreased soil invertebrate diversity even at label-recommended application rates. Critically, under the repeated-application regimes typical of conventional agriculture, negative effects on soil fauna did not attenuate: long-term field studies exhibited effect sizes comparable to short-term experiments [32, Sections 3.2–3.3, 4.2].

One of the most publicly visible consequences of the use of pesticides in industrial agriculture is the elevated loss rate of pollinator colonies. Neonicotinoid insecticides — the class most strongly implicated in bee population declines — are systemic compounds widely applied as prophylactic seed treatments, chronically exposing foraging pollinators through contaminated pollen and nectar [17, pp. 1255957-4–1255957-5]. Multiple interacting stressors — encompassing parasites, pathogens, nutritional deficiency, and pesticide exposure — represent the most probable drivers of elevated colony loss rates [17, p. 1255957-5]. Furthermore, the share of global agricultural area devoted to pollinator-dependent crops rose from 19.4% to 32.8% between 1961 and 2016, with expansion rates highest for crops in the moderate- and high-dependence categories [33, p. 3520], raising concern that continued pollinator decline could jeopardize the stability of agricultural food production.

The harm does not stop at the field boundary. A systematic review of epidemiological and toxicological evidence links chronic pesticide exposure to a wide spectrum of non-communicable diseases, including cancer, neurological disorders, and endocrine disruption [34, Sections 5–6]. A 2023 review of 63 epidemiological studies found that the strongest human evidence for associations between pesticide exposure and cancer concerned acute myeloid leukemia and colorectal cancer, where high-quality prospective studies observed dose-response relationships, though evidence for other cancer types remained inconsistent; the authors nonetheless concluded that existing evidence is sufficient to justify regulatory action [35, pp. 904–907]. Organochlorine and organophosphate compounds present in pesticides can mimic or antagonize endogenous hormones, with documented associations including reproductive disorders and developmental effects in children exposed *in utero* [34, Sections 5.1–5.2, 6.3]; these compounds have also been linked to breast and prostate cancers, though the epidemiological evidence remains mixed [35, pp. 900–901, 904].

The severity of these harms is not an unavoidable cost of pest management: it is in significant part a structural consequence of industrial open-field agriculture, where large-scale monoculture systems drive heightened pest pressure and perpetuate reliance on synthetic pesticides [36, p. 1].

2.6 Monoculture

Monoculture — the cultivation of a single crop species, or a narrow set of genetically near-identical cultivars, over large and continuous areas — is not merely one practice among many in industrial agriculture; it is its organizing principle. The economic logic of scale, mechanization, and commodity-market integration pushes production towards ever-greater uniformity, with cascading consequences for soil health, pest and disease dynamics, ecosystem resilience, and food-system stability. The following subsections document each of these dimensions in turn.

2.6.1 The monoculture effect: genetic vulnerability to disease and pest outbreaks

A fundamental result of epidemiological theory is that disease establishment requires host population density above a critical threshold [37, Section 1, pp. 535–536], and that genetically uniform host populations are disproportionately vulnerable to disease [38, pp. 381–382], [39, p. 718]. Monoculture maximizes both risk factors simultaneously: by planting a single crop species across large continuous areas, it concentrates the effective host population for any specialist pathogen or pest, while genetic uniformity ensures that a parasite capable of exploiting one individual can readily transmit to every neighbour, encountering no variation in resistance mechanisms.

This phenomenon — termed the "monoculture effect" — has been empirically quantified in a meta-analysis synthesizing 102 studies comprising over 2,000 effect sizes. In experimental crop systems, cultivar mixtures experienced approximately 50% less parasitism than monocultures on average — though the authors note this reflects a best-case scenario in which mixtures were assembled from cultivars with known resistance phenotypes [40, pp. 21, 23, 26].

The practical power of this effect has been demonstrated at operational scale. In a landmark field experiment in Yunnan Province, China, genetically diversified rice plantings across first five, then ten townships showed that disease-susceptible varieties planted in mixtures with resistant varieties suffered 94% less rice blast (the major disease of rice) severity than when grown in monoculture, and yielded 89% more per hill. The results were so dramatic that fungicidal sprays were no longer applied by the second year of the programme, though the authors note this may not be achievable in all seasons [39, pp. 718–721, Table 1]. This experiment remains one of the most compelling demonstrations that intraspecific crop diversification can function as a practical, large-scale disease management strategy.

The same vulnerability extends to insect pests. A review of 287 herbivore species found that approximately half (52%) reached higher population densities in monocultures, with only 15% showing the opposite pattern [41, Table 1, pp. 566–567], consistent with Root's resource concentration hypothesis — that herbivores arriving in concentrated host stands are more likely to remain and reproduce, driving specialist accumulation [42, p. 116]. A more recent global synthesis of 609 studies confirms that increasing plant species diversity consistently suppresses pest performance and promotes crop production across agro-ecosystems, grasslands, and forests[4] [43]. Natural enemy populations also tend to reach higher densities in diversified

[4] Cited from abstract only; full-text verification pending.

systems than in monocultures [41, Table 1, p. 567], reducing the biological control that would otherwise help regulate pest populations [44, p. 975]. The combination of elevated pest pressure and diminished biological control drives dependence on pesticides, whose application further disturbs the natural enemy communities that would limit outbreaks [44, pp. 975, 982], creating a self-reinforcing cycle.

2.6.2 Historical and contemporary case studies

The vulnerability inherent in genetic uniformity is not theoretical. It has repeatedly materialized in catastrophic losses throughout the history of modern agriculture.

The 1970 Southern Corn Leaf Blight—caused by a single cytoplasmic male-sterility trait (T-cytoplasm) shared by 75–90% of US hybrid maize cultivars [45, pp. 108–109]—destroyed an estimated 15% of the American crop in a single season and caused approximately $1 billion in losses [46, p. 1113] ($\geq$$6.0 billion in 2015 dollars [47, p. 1218]), a direct consequence of genetic homogeneity [46, p. 1113]. The disease had previously caused less than 1% average annual loss [46, p. 1115]; genetic uniformity converted a background pathogen into a national crisis in a single growing season. The lesson was not absorbed: by 2004, much of US corn germplasm still originated from just seven progenitor lines, and the tendency of breeders to recycle elite inbreds may, over the long term, erode the crop's genetic base [48, pp. 1193, 1203].

Global banana production depends overwhelmingly on a single cultivar — the Cavendish subgroup — which represents approximately 40% of global production and makes up almost all of the export trade, with an additional 20% of production consisting of closely related varieties sharing a very similar genetic background. The Cavendish itself rose to dominance only because its predecessor, the Gros Michel, was decimated by Fusarium wilt Race 1 from the 1920s onward — a disaster enabled by the genetic uniformity of the Gros Michel monoculture [49, pp. 2147, 2149–2150]. History is now repeating: a new strain, *Fusarium oxysporum* f.sp. *cubense* Tropical Race 4 (TR4), is spreading through Cavendish plantations worldwide: since symptoms were first observed in Cavendish in Taiwan in 1967 [49, p. 2150], TR4 has expanded to 23 countries as of 2024, including Colombia, Peru, and Venezuela [50, p. 1]. The pathogen can survive in soil for up to 30 years in the form of chlamydospores [51, p. 653], spreads through contaminated soil, water, and planting material, and there are currently no market-acceptable resistant cultivar replacements [52, p. 761]. More than 400 million people depend on bananas as a staple food or source of income, and the crop's combined economic value exceeds US$52 billion per year [49, pp. 2146, 2151]. The global banana industry thus faces an existential threat that is a direct structural consequence of monoculture production: an entire sector built on a single clone, with no viable fallback.

These are not isolated events. Coffee leaf rust — caused by *Hemileia vastatrix* — devastated Sri Lanka's coffee industry in the nineteenth century and remains a major threat to production worldwide [53, p. 1039]. The vulnerability is again structural: coffee is the main source of income for more than 100 million people [53, p. 1039], and *Coffea arabica*, which accounts for approximately 60% of world production, carries an exceptionally narrow genetic base — severe bottlenecks during domestication mean that a single coffee plant from the Botanical Garden of Amsterdam is believed to be one of the progenitors of most current cultivars [53, p. 1039]. A recent severe epidemic throughout Central America, Colombia, Peru, and Ecuador caused yield losses of up to 35%, directly affecting the livelihoods of hundreds of thousands of farmers [53,

p. 1040]. Wheat rust presents the same recurring pattern. Stem rust historically destroyed 19–28% of the US wheat crop in epidemic years from the 1910s to the 1950s [54, pp. 304–305]. A new threat emerged in 1998 with the Ug99 race of the stem rust fungus, which has since spread to 13 countries across Africa and the Middle East in 13 variants; the majority of wheat cultivars grown worldwide lack adequate resistance [54, pp. 305, 309]. Each of these cases — maize, banana, coffee, wheat — confirms the same structural lesson: production systems built on a narrow genetic base remain vulnerable to a single pathogen.

2.6.3 Soil degradation

Continuous monoculture degrades the very substrate upon which it depends. Growing the same crop repeatedly depletes specific nutrients, disrupts soil microbial communities and their functional capacity, and diminishes soil organic matter (SOM) and active carbon, which are essential to long-term soil productivity [55].

The impact on soil biota is also substantial. Monoculture systems exhibit reduced soil microbial diversity and altered community composition compared to diverse cropping systems [56]. The narrowing of root exudate chemistry under continuous cropping selects against a broad range of beneficial rhizosphere taxa — including disease-suppressive bacteria such as *Bacillus*, *Paenibacillus*, and *Lysinibacillus* — while favouring soil-borne pathogens such as *Fusarium* [57], [58]. Conversely, crop rotation increases microbial biomass carbon and nitrogen by around 13–16% and raises bacterial diversity relative to continuous monoculture [56], [59]. These biological improvements translate into measurable structural gains: a meta-analysis of 2,199 paired observations found that rotation improved macroaggregate stability[5] by 7–9% relative to continuous monoculture, with initial soil organic carbon identified as the predominant soil-level driver of this effect [61] — consistent with the established role of microbial biomass carbon and nitrogen in aggregate formation [62], and with the inverse losses that monoculture's suppression of carbon inputs and microbial communities imposes.

The resulting erosion is not a marginal effect. As documented previously (Section 2.3), erosion rates from conventionally ploughed monoculture fields average one to two orders of magnitude greater than rates of soil production — making the practice fundamentally unsustainable over civilisational timescales [19]. Where monocultures displace the diverse root architectures and ground cover of polycultures, rain runoff increases [63] and the soil's capacity to retain water diminishes [64], driving a secondary dependence on irrigation infrastructure [65].

2.6.4 Yield penalties and the diversification dividend

Far from maximizing output, monoculture imposes a persistent yield penalty relative to diversified systems. A global meta-analysis synthesizing 3,663 paired field-trial yield observations (1980–2024) found that crop rotation increased subsequent crop yield by 16–23% on average (depending on whether the pre-crop was a non-legume or legume), and that considering the entire rotation sequence, total yields, dietary energy,

[5] Macroaggregates are soil clumps >0.25 mm held together by organic matter and microbial binding agents. Their stability — resistance to breakdown by water and physical disturbance — governs erosion susceptibility, water infiltration, and the physical protection of organic carbon from decomposition [60], [61].

protein, iron, magnesium, zinc, and revenue all increased by 14–27% relative to continuous monoculture. Notably, win–win relationships among yield, nutrition, and revenue were consistently more frequent (33–54%) than trade-offs [66].

The benefits extend beyond yield quantity to yield stability — a dimension of critical importance for food security. An analysis of five decades of FAO data across 91 nations found that greater effective crop diversity at the national level was associated with significantly increased temporal stability of total national harvest. Countries with the lowest crop diversities experienced a severe food shortage (defined as a >25% decline in total yield) approximately every eight years; those with the highest diversities experienced one roughly every hundred years. The stabilizing effect of diversity was comparable in magnitude to the destabilizing effect of precipitation variability, indicating that crop diversification provides a quantitatively meaningful buffer against climatic shocks [67].

Long-term field experiments across North America further confirm that more diverse rotations increased maize yields in 7 of 11 sites during the most stressful growing conditions, with the largest resilience benefits observed where diversity differences were greatest [68]. This evidence is consistent with the broader ecological principle that diversity increases the stability and productivity of biological communities [69].

2.6.5 Diversification as a systemic alternative

The empirical case against monoculture is not limited to individual metrics. A second-order meta-analysis reviewing 98 first-order meta-analyses and encompassing 5,160 original studies (41,946 comparisons between diversified and simplified practices) found that agricultural diversification enhances biodiversity, pollination, pest control, nutrient cycling, soil fertility, and water regulation without compromising crop yields. In 63% of cases, diversification resulted in win–win outcomes for both ecosystem services and crop production simultaneously [70]. These results suggest that the simplification inherent in monoculture is not merely an environmental cost borne for the sake of production; it is a production strategy that is empirically dominated by more diverse alternatives across multiple dimensions.

An assessment of two agroindustrial cropping systems found that both had extremely low capacity to withstand climate and economic shocks, owing to low crop diversity, low landscape heterogeneity, and minimal vegetation cover — highlighting the structural fragility of monocropping in developing-country contexts [71].

2.6.6 Structural drivers: market concentration and monoculture

The persistence of monoculture despite its well-documented costs is not primarily a matter of agronomic ignorance; it is the structural outcome of concentrated market power and the economic incentives it creates. Four firms (BASF, Bayer, Corteva, and Syngenta) control 56% of the global commercial seed market and 61% of the global pesticides market — market concentrations that meet the standard economic definition of oligopoly [72]. Even where many farms exist, their effective genetic choices are constrained by a handful of input suppliers, making homogenisation not merely an agronomic preference but the structural outcome of concentrated market power. Consolidation drives a shift in research and development toward a small number of high-yielding proprietary varieties suited to monoculture systems and tied to proprietary agrochemical inputs. Farmers adopt these packages not merely out of preference but under structural compulsion: the agricultural treadmill[6] forces continuous yield maximisation simply to maintain income [73], while institutional lock-in — whereby research funding, career incentives, and scientific culture are organised around the dominant input-intensive paradigm, making it structurally difficult to develop, validate, or disseminate alternatives outside it — renders those alternatives inaccessible even where they demonstrably work [74]. The resulting genetic uniformity then creates the very pest and disease pressures established in Section 2.6.1, driving a self-reinforcing cycle of dependence on agrochemical inputs.

Concentrated market power enforces itself at the point of seed reproduction through two complementary strategies that prevent farmers from reproducing their own seed. The first is biological: hybrid varieties, whose offspring do not breed true, structurally eliminate the agronomic rationale for seed saving. The second is legal: the commercialization of transgenic seeds in the 1990s introduced full patent protections that prohibit seed saving outright, with violations subject to civil and criminal penalties [73]. The practical effect has been a rapid collapse of farm-level seed autonomy: the rate of saving soybean seed in the United States fell from 63% in 1960 to 10% by 2001 [73]. These strategies converge with bundling: patented seed traits are paired with proprietary agrochemical inputs in packages designed to foreclose substitution at each step, and as of 2009, the three then-largest seed firms already controlled 85% of transgenic corn patents and 70% of non-corn transgenic plant patents in the United States [73] — a concentration that subsequently deepened when, between 2015 and 2018, Bayer acquired Monsanto, Dow and DuPont merged to form Corteva Agrisciences, and ChemChina acquired Syngenta, reducing the sector's major players from six to four [75].

The concentration dynamics described above have been intensified in recent decades by the financialization of agrifood systems — the growing subordination of corporate strategy to the short-term return imperatives of financial markets — which has made mergers and acquisitions the preferred growth strategy for food and agricultural corporations, as firms pursue horizontal consolidation to generate shareholder value and capture market share. An analysis of 4,449 food system M&A deals between 2001 and 2020 found that the vast majority of activity was horizontal — firms acquiring competitors within the same sub-sector — accelerating concentration within sub-sectors rather than distributing it across them [76]. This process has been compounded by a parallel shift in ownership structure: large asset management firms have emerged as simultaneous major shareholders in dominant agrifood corporations across the sector, a pattern known as common ownership [77]. At their peak between 2010 and 2014, equity-related investment funds accounted for approximately one third of all financial investment in the agrifood sector [77]. Because firms with common owners are unlikely to compete aggressively against one another, this ownership structure adds a further anti-competitive layer to already concentrated markets [77]. The result is a system of mutually reinforcing structural forces — IP enforcement, merger-driven consolidation, and common ownership — that operates without any internal mechanism for self-correction.

[6]The agricultural treadmill, a concept introduced by economist Willard Cochrane in 1958, describes the mechanism by which farmers are compelled to continuously adopt productivity-increasing technologies: early adopters gain a temporary price advantage, but as adoption spreads, oversupply drives prices down, forcing all farmers to adopt the technology simply to maintain the same revenue. Those who cannot keep up exit farming; those who remain must adopt the next round of inputs. The result is an irreversible dependency on purchased inputs and a structural incentive to maximise yield per acre above all other considerations.

The structural character of these overlapping forces has a direct implication for the design of alternatives: technical solutions that operate within the same concentrated input markets will reproduce the same dependencies. Effective alternatives must be architecturally independent of proprietary input chains — which is to say, they must be open by design.

2.7 Freshwater depletion

Agriculture is the dominant consumer of global freshwater: while 70% of total withdrawal volume is the most widely cited estimate of its share, a 2025 uncertainty analysis drawing on eleven global hydrological models and thirteen independent datasets places the 95% credible range at 70–85%, suggesting the true share is at least as high as the commonly reported figure and likely higher [78].

The full scale of agricultural water consumption is partially obscured because it includes the depletion of non-renewable aquifers. Global groundwater depletion increased 22% in just one decade, from 240 km^3/year in 2000 to 292 km^3/year in 2010, with five crops — wheat (22% of global GWD), rice (17%), sugar crops (7%), cotton (7%), and maize (5%) — dominating depletion in production, and with all traded agricultural commodities combined accounting for 83% of the global GWD total [79]. Unlike surface water shortages, depletion of fossil aquifers is functionally irreversible on human timescales: some of the groundwater currently consumed to irrigate these crops fell as precipitation thousands of years ago and will not recharge within any planning horizon relevant to food security policy [80].

2.8 Greenhouse gas emissions

The greenhouse gas contribution of the global agrifood system can be quantified through complementary accounting frameworks that differ primarily in system boundary.[7] The FAOSTAT boundary — covering farm-gate[8] production, land-use change, and pre- and post-production supply-chain activities directly attributable to the agrifood sector — places global emissions at 16.2 Gt CO_2eq for 2022, equivalent to approximately 30% of total anthropogenic GHG emissions [81]. The EDGAR-FOOD database, which applies a broader lifecycle boundary that includes upstream fertiliser and pesticide manufacture and downstream household preparation and landfill, estimated 18 Gt CO_2eq and 34% for 2015 [82]. The IPCC's Special Report on Climate Change and Land, drawing on multiple estimation methods, places the full food-system contribution at 21–37% of total net anthropogenic emissions for 2007–2016 [21, SPM A.3]. Within the FAOSTAT framework, emissions divide across three domains: farm-gate activities (48%; 7.8 Gt), land-use change (19%; 3.1 Gt), and the supply chain (33%; 5.3 Gt) [81].

The dominant source of food-system emissions is the production of animal-based food (including livestock feed), which accounts for 57% of total food-related GHG emissions [83] while providing only 37% of global protein and 18% of calories [84] — a disproportionality whose implications for dietary strategy are examined in Section 2.11.

[7] A *system boundary* defines which activities along the food chain are counted as emission sources — e.g. whether only on-farm processes are included or also upstream manufacturing and downstream retail.

[8] *Farm-gate* emissions comprise all GHG releases from crop and livestock production activities within the agricultural holding, up to the point the product leaves the farm.

The agrifood sector accounts for a structurally disproportionate share of the most potent non-CO_2 greenhouse gases. The IPCC estimates that agriculture, forestry, and other land use (AFOLU) is responsible for approximately 44% of all anthropogenic methane and 81% of all anthropogenic nitrous oxide globally [21, SPM A.3]. The N_2O trajectory — its 40% growth since 1980, exceedance of all CMIP6 emission scenarios, and the reduction required for 1.5°C compatibility — is examined in detail in the preceding discussion of fertilizer use [29]. Methane is of particular strategic importance because its atmospheric perturbation lifetime is only 11.8 ± 1.8 years [28, Table 7.15], compared with over a century for nitrous oxide and millennia for CO_2. A sustained reduction in methane emissions therefore translates into measurable cooling within two decades [28, Section 7.6.1.4]. Since ruminants[9] and rice cultivation are the dominant agricultural sources of methane [21, SPM A.3.4], shifting food production away from ruminant-dependent systems represents one of the most actionable near-term levers for slowing global warming.

The structural implication of these trajectories is severe. Even if all fossil fuel combustion were halted immediately, food-system emissions continuing on a business-as-usual path would alone exhaust the remaining carbon budget for 1.5°C between 2051 and 2063, and would consume over 96% of the budget consistent with a 67% chance of limiting warming to 2°C by end of century[10] [85]. Meeting Paris Agreement targets therefore requires the simultaneous implementation of multiple food-system interventions — dietary shifts, caloric adjustment, yield improvements, waste reduction, and production efficiency gains — with no single strategy sufficient. The agrifood system is therefore not a peripheral contributor to the climate crisis but a structural driver of it, whose reform is a precondition of any plausible stabilisation pathway [85].

2.9 Produce transportation, storage and disposal

During 2010–2016, global food loss and waste accounted for an estimated 8–10% of total anthropogenic greenhouse gas emissions. An estimated 25–30% of all food produced is lost or wasted[11], with losses in developing countries driven by inadequate infrastructure—such as lack of refrigeration—while waste in developed countries is driven by consumer behaviour [21, Section 5.5.2.5].

Beyond the farm gate, food-system emissions from transport, retail, and household consumption have grown substantially. Crippa et al. estimate that food transport contributes 4.8% and retail 4.0% of total food-system emissions, with transport's share of food-system emissions rising by 67% and retail's share by 300% between 1990 and 2015 [82, pp. 200–201]. More recent data show that pre- and post-production emissions reached 5.3 Gt CO_2eq in 2022—33% of all agrifood emissions, up from 24% in 2000—with all pre- and post-production subcategories—manufacturing of inputs, transport, packaging and retail, and household consumption—each growing by more than 80% over the same period [81, pp. 3–4]. Cold chain infrastructure alone—refrigeration across processing, transport, retail, and household stages—accounts for an estimated

[9] Ruminants are herbivorous mammals — principally cattle, sheep, and goats — that digest plant matter through enteric fermentation in a specialised multi-chambered stomach, a process that releases methane as a metabolic by-product.

[10] Clark et al. report emissions in CO_2 warming-equivalents (CO_2-we), a metric based on GWP* that captures the different temporal dynamics of short-lived (CH_4) and long-lived (CO_2, N_2O) climate forcers. Cumulative budgets expressed in the more common GWP_{100}-based CO_2eq would yield different numerical thresholds.

[11] The IPCC assigns this estimate *medium confidence* (SRCCL, Section 5.5.2.5).

1.32 Gt CO_2eq[12], or nearly 10% of total agrifood emissions, having more than doubled since 2000 [86, Section 3]. In 61 countries and territories, supply-chain emissions now account for 60% or more of national agrifood emissions [81, p. 7].

Food waste that reaches landfills is a disproportionate source of methane. In the United States, food waste constitutes an estimated 24% of municipal solid waste disposed of in landfills, yet owing to its rapid decay rate, an estimated 58% of fugitive methane emissions from municipal solid waste landfills are attributable to landfilled food waste [87, Table A1, Table 3][13]. Food waste decomposes faster than other organic materials, and an estimated 61% of the methane it generates escapes to the atmosphere uncaptured [87, p. iii].

2.10 The existential dependency on a few producers

The global food system is vulnerable to disruption at multiple levels of concentration. Three global food crises in the past fifty years—1973–1974, 2007–2012, and the 2019–2022 pandemic and conflict disruptions—have each exposed interlocking structural weaknesses: the system's reliance on a small number of staple grains produced using highly industrialised methods, the geographic concentration of the countries that export them, and the dominance of a small number of firms over global commodity markets [88, pp. 1–2]. In 2021–2022, seven countries plus the European Union accounted for 84% of world wheat exports, and four countries accounted for 83% of world maize exports; by 2024–2025, wheat concentration had risen to 88% [89, pp. 16, 25]. The majority of nations do not produce enough staple grains to meet domestic demand and therefore depend on global markets dominated by these few exporters [88, p. 6].

The concentration risk extends upstream. The oligopolistic control of the commercial seed and pesticide markets documented in Section 2.6.6 is mirrored in agricultural machinery, where the top four firms control 43% of the global market [72, Table 4]. This degree of corporate concentration has been linked to reduced competition—though empirical evidence on pricing effects remains limited by restricted access to proprietary industry data—and to narrower technological choices, as dominant firms focus on high-cost proprietary input packages over less capital-intensive alternatives [90, pp. 404–406].

At the most local level, food production itself is consolidating into fewer and larger operations. Globally, 0.1% of farms—those exceeding 1 000 hectares—operate over half of all agricultural land, while the 85% of farms smaller than 2 hectares operate just 9% [91, Chapter 3]. Farms exceeding 50 hectares produce more than 55% of crop-derived dietary energy worldwide[14] [91, Chapter 3]. If historical patterns of structural transformation continue, the total number of farms worldwide is projected to decrease by half before the end of the century—even as farm numbers in sub-Saharan Africa are expected to rise [91, Chapter 3]. Smallholders—farms under 2 hectares, which still produce 16% of global crop-derived dietary energy—are vital contributors to food supply, yet the prevailing trajectory of consolidation is reducing their number [91, Chapter 3]. The result is a food system in which the vast majority of people—including entire nations—depend for their sustenance on supply chains they neither control nor can easily replace.

[12]Flammini et al. report 30–50% uncertainty on this estimate due to limitations in energy-use and refrigerant-leakage data.

[13]These figures are specific to the United States. No comparable global analysis of food waste's share of landfill methane emissions exists in the peer-reviewed literature as of early 2026.

[14]This estimate is based on crop production data from 77 countries covering approximately 54% of global cropland area (FAO SOFA 2025, Chapter 3).

2.11 Suppression of dietary diversity and nutritional quality

Industrialised agriculture prioritises monoculture production of a narrow range of commodity crops, the cheap extracted ingredients of which—oils, starches, sugars—form the basis of ultra-processed foods[15] that now dominate the food supplies of high-income countries and are rising rapidly in middle-income countries [93, pp. 21, 27] [92, Figure 1]. An umbrella review of 14 meta-analyses encompassing nearly 10 million participants found convincing evidence that greater ultra-processed food exposure is associated with higher risks of cardiovascular disease mortality and type 2 diabetes, and highly suggestive evidence of association with obesity[16] [94, Figures 2–3].

The nutritional cost of industrial agriculture extends beyond dietary composition to the crops themselves. Analyses of archived wheat grain from the Broadbalk long-term experiment (1845–2005) show that zinc, copper, and magnesium concentrations declined significantly after the mid-1960s—by 33–49% for zinc, 25–39% for copper, and 20–27% for magnesium—coinciding with the introduction of semi-dwarf high-yield cultivars. Iron concentrations were also significantly lower in short-straw cultivars than in their long-straw predecessors, though the within-period temporal trend was not statistically significant [95, Tables 2, 4]. Soil mineral levels remained stable or increased over the same period, indicating that the decline is a plant-level phenomenon rather than a consequence of soil depletion [95, p. 321]. A broader study of 43 garden crops using USDA composition data from 1950 to 1999 found statistically reliable declines in protein (−6%), calcium (−16%), iron (−15%), and riboflavin (−38%), among others, attributing the pattern to a "dilution effect" in which selection for higher yields reduces the concentration of minerals and other nutrients per unit of harvested mass [96, Table 3]. Similar trends have been documented in UK fruits and vegetables over an 80-year period, with iron declining by approximately 50% and copper by 49%[17] [98, Table 1]. These declines are consequential because iron, zinc, and vitamin A deficiencies constitute the most prevalent forms of micronutrient malnutrition globally, with the heaviest burden falling on populations that depend on staple cereal and legume crops for the bulk of their dietary intake [99, pp. 353–354]; the documented decline in mineral density of precisely these crops compounds an already severe public health problem.

[15] Ultra-processed foods are defined under the NOVA classification as branded, commercial formulations made from cheap ingredients extracted or derived from whole foods and combined with additives, most containing little to no whole food [92, Panel 1].

[16] All evidence on ultra-processed food and long-term health outcomes is necessarily observational—randomising participants to decades of dietary exposure is not ethically feasible. Lane et al. report the highest available tier of evidence synthesis for this question; their "convincing" (Class I) and "highly suggestive" (Class II) grades reflect the strength of association within that framework, not experimental proof of causation. The underlying GRADE quality ratings were predominantly low or very low (only 4 of 45 pooled analyses reached moderate; none reached high). Under GRADE methodology, observational studies are rated "low" quality by default, so these ratings reflect inherent limits of the study design rather than weakness in the observed associations.

[17] Nutrient-decline estimates derived from historical food-composition tables are contested. A critical review by Marles [97, Sections 3–4, 6–7] argues that changes in analytical methods, sampling protocols, and cultivar selection across editions render direct comparisons unreliable. However, Marles concedes that the archived-sample evidence of Fan et al. avoids these methodological limitations and that the dilution effect from high-yield cultivar selection is a real phenomenon, though he argues that its magnitude is too small to have a significant impact on consumer nutrition and that increased yield outweighs the dilution cost. The estimates from Davis et al. and Mayer et al. should therefore be read as indicative of direction and approximate magnitude, not as precise measurements.

Beyond nutrient decline, the narrowing of crop diversity documented above has also homogenised food supplies at a global scale. An analysis of national food supply data from 152 countries over 50 years found that national food supplies converged significantly towards a common pattern dominated by a few staple cereals and oil crops, even as the number of available crop commodities increased—a convergence in proportional composition rather than in crop availability—notably wheat, rice, maize, soybean, and palm oil, with regions whose food supplies were most compositionally distinct—such as East and Southeast Asia and sub-Saharan Africa—experiencing the most rapid convergence [100, Figure 3]. Modelling by Springmann et al. shows that current dietary trends — projected increases in consumption of animal products, sugar, and oils driven by population and income growth — could, in the absence of technological changes and dedicated mitigation measures, push multiple planetary boundaries—biophysical thresholds beyond which Earth-system processes risk destabilisation—beyond safe limits by 2050, including those for climate change, cropland use, freshwater use, and nitrogen and phosphorus pollution [101, p. 522]—reinforcing the conclusion, noted in Section 2.8, that no single intervention is sufficient and that systemic reform of the food system is required.

References

[1] Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services, "Global assessment report of the intergovernmental science-policy platform on biodiversity and ecosystem services," IPBES Secretariat, Bonn, Germany, Tech. Rep., 2019, p. 1144. DOI: 10.5281/zenodo.3831673.

[2] European Commission: Directorate-General for Financial Stability, Financial Services and Capital Markets Union, IIASA & Trinomics, *Study for a methodological framework and assessment of potential financial risks associated with biodiversity loss and ecosystem degradation – Final report*. Publications Office of the European Union, 2024. DOI: 10.2874/30768. [Online]. Available: https://data.europa.eu/doi/10.2874/30768.

[3] L. Cabernard *et al.*, "Biodiversity impacts of recent land-use change driven by increases in agri-food imports," *Nature Sustainability*, 2024. DOI: 10.1038/s41893-024-01433-4.

[4] Food and Agriculture Organization of the United Nations, "Land statistics 2001–2023 – global, regional and country trends," FAOSTAT Analytical Briefs, Rome, Tech. Rep. No. 107, 2025, FAO. DOI: 10.4060/cd5765en. [Online]. Available: https://openknowledge.fao.org/handle/20.500.14283/cd5765en.

[5] E. S. Cassidy, P. C. West, J. S. Gerber, and J. A. Foley, "Redefining agricultural yields: From tonnes to people nourished per hectare," *Environmental Research Letters*, vol. 8, no. 3, p. 034 015, 2013. DOI: 10.1088/1748-9326/8/3/034015.

[6] Food and Agriculture Organization of the United Nations, "The state of food and agriculture 2025 – addressing land degradation across landholding scales," FAO, Rome, Tech. Rep., 2025, FAO publication. DOI: 10.4060/cd7067en. [Online]. Available: https://doi.org/10.4060/cd7067en.

[7] D. Lawrence, M. Coe, W. Walker, L. Verchot, and K. Vandecar, "The unseen effects of deforestation: Biophysical effects on climate," *Front. For. Glob. Change*, vol. 5, Mar. 2022. DOI: 10.3389/ffgc.2022.756115. [Online]. Available: https://www.frontiersin.org/journals/forests-and-global-change/articles/10.3389/ffgc.2022.756115/full.

[8] F. Pendrill, T. A. Gardner, P. Meyfroidt, U. M. Persson, *et al.*, "Disentangling the numbers behind agriculture-driven tropical deforestation," *Science*, vol. 377, no. 6611, eabm9267, 2022. DOI: 10.1126/science.abm9267.

[9] P. H. Raven and D. L. Wagner, "Agricultural intensification and climate change are rapidly decreasing insect biodiversity," *Proceedings of the National Academy of Sciences*, vol. 118, no. 2, 2021. DOI: 10.1073/pnas.2002548117.

[10] T. G. Benton, C. Bieg, H. Harwatt, R. Pudasaini, and L. Wellesley, "Food system impacts on biodiversity loss: Three levers for food system transformation in support of nature," Royal Institute of International Affairs, 2021. [Online]. Available: https://www.chathamhouse.org/2021/02/food-system-impacts-biodiversity-loss.

[11] C. A. Hallmann, M. Sorg, E. Jongejans, *et al.*, "More than 75 percent decline over 27 years in total flying insect biomass in protected areas," *PLOS ONE*, vol. 12, no. 10, e0185809, 2017. DOI: 10.1371/journal.pone.0185809.

[12] D. L. Wagner, E. M. Grames, M. L. Forister, M. R. Berenbaum, and D. Stopak, "Insect decline in the Anthropocene: Death by a thousand cuts," *Proceedings of the National Academy of Sciences*, vol. 118, no. 2, e2023989118, 2021. DOI: 10.1073/pnas.2023989118.

[13] R. van Klink, D. E. Bowler, K. B. Gongalsky, M. Shen, S. R. Swengel, and J. M. Chase, "Disproportionate declines of formerly abundant species underlie insect loss," *Nature*, vol. 628, pp. 359–364, 2024. DOI: 10.1038/s41586-023-06861-4.

[14] C. L. Outhwaite, P. McCann, and T. Newbold, "Agriculture and climate change are reshaping insect biodiversity worldwide," *Nature*, vol. 605, pp. 97–102, 2022. DOI: 10.1038/s41586-022-04644-x.

[15] D. M. Salcido, M. L. Forister, H. Garcia Lopez, and L. A. Dyer, "Loss of dominant caterpillar genera in a protected tropical forest," *Scientific Reports*, vol. 10, p. 422, 2020. DOI: 10.1038/s41598-019-57226-9.

[16] S. Rigal, V. Dakos, H. Alonso, *et al.*, "Farmland practices are driving bird population decline across Europe," *Proceedings of the National Academy of Sciences*, vol. 120, no. 21, e2216573120, 2023. DOI: 10.1073/pnas.2216573120.

[17] D. Goulson, E. Nicholls, C. Botías, and E. L. Rotheray, "Bee declines driven by combined stress from parasites, pesticides, and lack of flowers," *Science*, vol. 347, no. 6229, p. 1 255 957, 2015. DOI: 10.1126/science.1255957.

[18] Food and Agriculture Organization of the United Nations, *Revised World Soil Charter*, en. Rome, Italy: FAO, 2015. [Online]. Available: https://openknowledge.fao.org/server/api/core/bitstreams/65be618a-f5e8-4f67-9cbb-d77f1f5cd31e/content.

[19] D. R. Montgomery, "Soil erosion and agricultural sustainability," *Proceedings of the National Academy of Sciences*, vol. 104, no. 33, pp. 13 268–13 272, 2007. DOI: 10.1073/pnas.0611508104.

[20] R. Lal, "Soil carbon sequestration to mitigate climate change," *Geoderma*, vol. 123, no. 1–2, pp. 1–22, 2004. DOI: 10.1016/j.geoderma.2004.01.032.

[21] IPCC, "Climate change and land: An IPCC special report on climate change, desertification, land degradation, sustainable land management, food security, and greenhouse gas fluxes in terrestrial ecosystems," Intergovernmental Panel on Climate Change, Cambridge, UK and New York, NY, USA, Tech. Rep., 2019. DOI: 10.1017/9781009157988. [Online]. Available: https://www.ipcc.ch/srccl/.

[22] D. Beillouin *et al.*, "A global meta-analysis of soil organic carbon in the Anthropocene," *Nature Communications*, vol. 14, p. 3700, 2023. DOI: 10.1038/s41467-023-39338-z.

[23] M. F. Chislock, E. Doster, R. A. Zitomer, and A. E. Wilson, "Eutrophication: Causes, Consequences, and Controls in Aquatic Ecosystems," *Nature Education Knowledge*, vol. 4, no. 4, p. 10, 2013. [Online]. Available: https://www.nature.com/scitable/knowledge/library/eutrophication-causes-consequences-and-controls-in-aquatic-102364466.

[24] S. R. Carpenter, N. F. Caraco, D. L. Correll, R. W. Howarth, A. N. Sharpley, and V. H. Smith, "Nonpoint pollution of surface waters with phosphorus and nitrogen," *Issues in Ecology*, no. 3, 1998. [Online]. Available: https://esa.org/wp-content/uploads/2013/03/issue3.pdf.

[25] R. J. Diaz and R. Rosenberg, "Spreading dead zones and consequences for marine ecosystems," *Science*, vol. 321, no. 5891, pp. 926–929, 2008. DOI: 10.1126/science.1156401.

[26] D. Breitburg, L. A. Levin, A. Oschlies, *et al.*, "Declining oxygen in the global ocean and coastal waters," *Science*, vol. 359, no. 6371, eaam7240, 2018. DOI: 10.1126/science.aam7240.

[27] D. Coskun, D. T. Britto, W. Shi, and H. J. Kronzucker, "Nitrogen transformations in modern agriculture and the role of biological nitrification inhibition," *Nature Plants*, vol. 3, p. 17 074, 2017. DOI: 10.1038/nplants.2017.74.

[28] P. Forster, T. Storelvmo, K. Armour, *et al.*, "The earth's energy budget, climate feedbacks, and climate sensitivity," in *Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change*, V. Masson-Delmotte, P. Zhai, A. Pirani, *et al.*, Eds., Cambridge, United Kingdom and New York, NY, USA: Cambridge University Press, 2021, ch. 7, pp. 923–1054. DOI: 10.1017/9781009157896.009.

[29] H. Tian, N. Pan, R. L. Thompson, *et al.*, "Global nitrous oxide budget (1980–2020)," *Earth System Science Data*, vol. 16, no. 6, pp. 2543–2604, 2024. DOI: 10.5194/essd-16-2543-2024.

[30] I. Shcherbak, N. Millar, and G. P. Robertson, "Global metaanalysis of the nonlinear response of soil nitrous oxide (n2o) emissions to fertilizer nitrogen," *Proceedings of the National Academy of Sciences*, vol. 111, no. 25, pp. 9199–9204, 2014. DOI: 10.1073/pnas.1322434111. eprint: https://www.pnas.org/doi/pdf/10.1073/pnas.1322434111. [Online]. Available: https://www.pnas.org/doi/abs/10.1073/pnas.1322434111.

[31] F. H. M. Tang, M. Lenzen, A. McBratney, and F. Maggi, "Risk of pesticide pollution at the global scale," *Nature Geoscience*, vol. 14, pp. 206–210, Apr. 2021. DOI: 10.1038/s41561-021-00712-5.

[32] L. Beaumelle, L. Tison, N. Eisenhauer, *et al.*, "Pesticide effects on soil fauna communities—A meta-analysis," *Journal of Applied Ecology*, vol. 60, no. 7, pp. 1239–1253, 2023. DOI: 10.1111/1365-2664.14437.

[33] M. A. Aizen, S. Aguiar, J. C. Biesmeijer, *et al.*, "Global agricultural productivity is threatened by increasing pollinator dependence without a parallel increase in crop diversification," *Global Change Biology*, vol. 25, no. 10, pp. 3516–3527, 2019. DOI: 10.1111/gcb.14736.

[34] C. Shekhar, R. Khosya, K. Thakur, *et al.*, "A systematic review of pesticide exposure, associated risks, and long-term human health impacts," *Toxicology Reports*, vol. 13, p. 101 840, 2024, ISSN: 2214-7500. DOI: 10.1016/j.toxrep.2024.101840. [Online]. Available: https://www.sciencedirect.com/science/article/pii/S2214750024002233.

[35] H. Cavalier, L. Trasande, and M. Porta, "Exposures to pesticides and risk of cancer: Evaluation of recent epidemiological evidence in humans and paths forward," *International Journal of Cancer*, vol. 152, no. 5, pp. 879–912, 2023. DOI: 10.1002/ijc.34300.

[36] S. R. Belmain, Y. Tembo, A. G. Mkindi, S. E. J. Arnold, and P. C. Stevenson, "Elements of agroecological pest and disease management," *Elementa: Science of the Anthropocene*, vol. 10, no. 1, p. 00 099, 2022. DOI: 10.1525/elementa.2021.00099.

[37] R. M. Anderson and R. M. May, "The invasion, persistence and spread of infectious diseases within animal and plant communities," *Philosophical Transactions of the Royal Society of London. Series B, Biological Sciences*, vol. 314, no. 1167, pp. 533–570, 1986. DOI: 10.1098/rstb.1986.0072.

[38] C. C. Mundt, "Use of multiline cultivars and cultivar mixtures for disease management," *Annual Review of Phytopathology*, vol. 40, pp. 381–410, 2002. DOI: 10.1146/annurev.phyto.40.011402.113723.

[39] Y. Zhu, H. Chen, J. Fan, *et al.*, "Genetic diversity and disease control in rice," *Nature*, vol. 406, pp. 718–722, 2000. DOI: 10.1038/35021046.

[40] A. K. Gibson and A. E. Nguyen, "Does genetic diversity protect host populations from parasites? A meta-analysis across natural and agricultural systems," *Evolution Letters*, vol. 5, no. 1, pp. 16–32, 2021. DOI: 10.1002/evl3.206.

[41] D. A. Andow, "Vegetational diversity and arthropod population response," *Annual Review of Entomology*, vol. 36, no. 1, pp. 561–586, 1991. DOI: 10.1146/annurev.en.36.010191.003021.

[42] R. B. Root, "Organization of a plant-arthropod association in simple and diverse habitats: The fauna of collards (*Brassica oleracea*)," *Ecological Monographs*, vol. 43, no. 1, pp. 95–124, 1973. DOI: 10.2307/1942161.

[43] N.-F. Wan, Y.-Q. Wang, L. Fu, *et al.*, "Global evidence that plant diversity suppresses pests and promotes plant performance and crop production," *Nature Ecology & Evolution*, 2025. DOI: 10.1038/s41559-025-02964-5.

[44] J. F. Tooker and S. D. Frank, "Genotypically diverse cultivar mixtures for insect pest management and increased crop yields," *Journal of Applied Ecology*, vol. 49, no. 5, pp. 974–985, 2012. DOI: 10.1111/j.1365-2664.2012.02173.x.

[45] National Research Council, *Genetic Vulnerability of Major Crops*. Washington, DC: The National Academies Press, 1972. DOI: 10.17226/27600.

[46] L. A. Tatum, "The southern corn leaf blight epidemic," *Science*, vol. 171, no. 3976, pp. 1113–1116, 1971. DOI: 10.1126/science.171.3976.1113.

[47] H. A. Bruns, "Southern corn leaf blight: A story worth retelling," *Agronomy Journal*, vol. 109, no. 4, pp. 1218–1224, 2017. DOI: 10.2134/agronj2017.01.0006.

[48] M. A. Mikel and J. W. Dudley, "Evolution of North American dent corn from public to proprietary germplasm," *Crop Science*, vol. 46, no. 3, pp. 1193–1205, 2006. DOI: 10.2135/cropsci2005.10-0371.

[49] A. Drenth and G. Kema, "The vulnerability of bananas to globally emerging disease threats," *Phytopathology*, vol. 111, no. 12, pp. 2146–2161, 2021, PMID: 34231377. DOI: 10.1094/PHYTO-07-20-0311-RVW.

[50] T. Munhoz, J. Vargas, L. Teixeira, C. Staver, and M. Dita, "Fusarium Tropical Race 4 in Latin America and the Caribbean: Status and global research advances towards disease management," *Frontiers in Plant Science*, vol. 15, p. 1 397 617, 2024. DOI: 10.3389/fpls.2024.1397617.

[51] R. C. Ploetz, "Fusarium wilt of banana is caused by several pathogens referred to as *Fusarium oxysporum* f. sp. *cubense*," *Phytopathology*, vol. 96, no. 6, pp. 653–656, 2006. DOI: 10.1094/PHYTO-96-0653.

[52] J. M. Roberts, L. C. Carvalhais, C. O'Dwyer, V. A. Rincón-Flórez, and A. Drenth, "Diagnostics of Fusarium wilt in banana: Current status and challenges," *Plant Pathology*, vol. 73, no. 4, pp. 760–776, 2024. DOI: 10.1111/ppa.13863.

[53] P. Talhinhas, D. Batista, I. Diniz, *et al.*, "The coffee leaf rust pathogen *Hemileia vastatrix*: One and a half centuries around the tropics," *Molecular Plant Pathology*, vol. 18, no. 8, pp. 1039–1051, 2017. DOI: 10.1111/mpp.12512.

[54] R. P. Singh, P. K. Singh, J. Rutkoski, *et al.*, "Disease impact on wheat yield potential and prospects of genetic control," *Annual Review of Phytopathology*, vol. 54, pp. 303–322, 2016. DOI: 10.1146/annurev-phyto-080615-095835.

[55] Z. H. Pervaiz, J. Iqbal, Q. Zhang, D. Chen, H. Wei, and M. Saleem, "Continuous cropping alters multiple biotic and abiotic indicators of soil health," *Soil Systems*, vol. 4, no. 4, p. 59, 2020. DOI: 10.3390/soilsystems4040059. [Online]. Available: https://doi.org/10.3390/soilsystems4040059.

[56] C. Li, L. Shi, K. Wang, *et al.*, "Crop rotation differentially increases soil bacterial and fungal diversities in global croplands: A meta-analysis," *Nature Communications*, vol. 16, p. 11 686, 2025. DOI: 10.1038/s41467-025-66823-4.

[57] P. Li, J. Liu, M. Saleem, *et al.*, "Reduced chemodiversity suppresses rhizosphere microbiome functioning in the mono-cropped agroecosystems," *Microbiome*, vol. 10, p. 108, 2022. DOI: 10.1186/s40168-022-01287-y.

[58] Y. Zhou, Z. Yang, J. Liu, *et al.*, "Crop rotation and native microbiome inoculation restore soil capacity to suppress a root disease," *Nature Communications*, vol. 14, no. 1, p. 8126, 2023. DOI: 10.1038/s41467-023-43926-4.

[59] Q. Liu, Y. Zhao, T. Li, L. Chen, Y. Chen, and P. Sui, "Changes in soil microbial biomass, diversity, and activity with crop rotation in cropping systems: A global synthesis," *Applied Soil Ecology*, vol. 186, p. 104 815, 2023. DOI: 10.1016/j.apsoil.2023.104815.

[60] J. M. Tisdall and J. M. Oades, "Organic matter and water-stable aggregates in soils," *Journal of Soil Science*, vol. 33, no. 2, pp. 141–163, 1982. DOI: 10.1111/j.1365-2389.1982.tb01755.x.

[61] F. Zheng, X. Liu, W. Ding, X. Song, S. Li, and X. Wu, "Positive effects of crop rotation on soil aggregation and associated organic carbon are mainly controlled by climate and initial soil carbon content: A meta-analysis," *Agriculture, Ecosystems & Environment*, vol. 355, p. 108 600, 2023. DOI: 10.1016/j.agee.2023.108600.

[62] M. D. McDaniel, L. K. Tiemann, and A. S. Grandy, "Does agricultural crop diversity enhance soil microbial biomass and organic matter dynamics? A meta-analysis," *Ecological Applications*, vol. 24, no. 3, pp. 560–570, 2014. DOI: 10.1890/13-0616.1.

[63] A. D. Basche and M. S. DeLonge, "Comparing infiltration rates in soils managed with conventional and alternative farming methods: A meta-analysis," *PLOS ONE*, vol. 14, no. 9, e0215702, 2019. DOI: 10.1371/journal.pone.0215702.

[64] A. D. Basche and M. DeLonge, "The impact of continuous living cover on soil hydrologic properties: A meta-analysis," *Soil Science Society of America Journal*, vol. 81, no. 5, pp. 1179–1190, 2017. DOI: 10.2136/sssaj2017.03.0077.

[65] L. Alletto, A. Vandewalle, and P. Debaeke, "Crop diversification improves cropping system sustainability: An 8-year on-farm experiment in south-western france," *Agricultural Systems*, vol. 200, p. 103 433, 2022. DOI: 10.1016/j.agsy.2022.103433.

[66] S. Mudare, J. Jing, D. Makowski, *et al.*, "Crop rotations synergize yield, nutrition, and revenue: A meta-analysis," *Nature Communications*, vol. 16, p. 9552, 2025. DOI: 10.1038/s41467-025-64567-9.

[67] D. Renard and D. Tilman, "National food production stabilized by crop diversity," *Nature*, vol. 571, no. 7764, pp. 257–260, 2019. DOI: 10.1038/s41586-019-1316-y.

[68] T. M. Bowles, M. Mooshammer, Y. Socolar, *et al.*, "Long-term evidence shows that crop-rotation diversification increases agricultural resilience to adverse growing conditions in North America," *One Earth*, vol. 2, no. 3, pp. 284–293, 2020. DOI: 10.1016/j.oneear.2020.02.007.

[69] B. J. Cardinale, J. E. Duffy, A. Gonzalez, *et al.*, "Biodiversity loss and its impact on humanity," *Nature*, vol. 486, no. 7401, pp. 59–67, 2012. DOI: 10.1038/nature11148.

[70] G. Tamburini, R. Bommarco, T. C. Wanger, *et al.*, "Agricultural diversification promotes multiple ecosystem services without compromising yield," *Science Advances*, vol. 6, no. 45, eaba1715, 2020. DOI: 10.1126/sciadv.aba1715.

[71] S. Mukhovi and J. Jacobi, "Can monocultures be resilient? assessment of buffer capacity in two agroindustrial cropping systems in africa and south america," *Agriculture & Food Security*, vol. 11, no. 19, 2022. DOI: 10.1186/s40066-022-00356-7.

[72] ETC Group and GRAIN, "Top 10 agribusiness giants: Corporate concentration in food & farming in 2025," ETC Group and GRAIN, Tech. Rep., 2025. [Online]. Available: https://www.etcgroup.org/content/top-10-agribusiness-giants.

[73] P. H. Howard, "Visualizing consolidation in the global seed industry: 1996–2008," *Sustainability*, vol. 1, no. 4, pp. 1266–1287, 2009. DOI: 10.3390/su1041266.

[74] G. Vanloqueren and P. V. Baret, "How agricultural research systems shape a technological regime that develops genetic engineering but locks out agroecological innovations," *Research Policy*, vol. 38, no. 6, pp. 971–983, 2009. DOI: 10.1016/j.respol.2009.02.008.

[75] K. Deconinck, "Concentration in seed and biotech markets: Extent, causes, and impacts," *Annual Review of Resource Economics*, vol. 12, pp. 129–147, 2020. DOI: 10.1146/annurev-resource-102319-100751.

[76] L. Keenan, T. Monteath, and D. Wójcik, "Hungry for power: Financialization and the concentration of corporate control in the global food system," *Geoforum*, vol. 147, p. 103 909, 2023. DOI: 10.1016/j.geoforum.2023.103909.

[77] J. Clapp, "The rise of financial investment and common ownership in global agrifood firms," *Review of International Political Economy*, vol. 26, no. 4, pp. 604–629, 2019. DOI: 10.1080/09692290.2019.1597755.

[78] A. Puy, S. N. Linga, N. Wei, *et al.*, "Widely cited global irrigation statistics lack empirical support," *PNAS Nexus*, vol. 4, no. 11, pgaf323, 2025. DOI: 10.1093/pnasnexus/pgaf323.

[79] C. Dalin, Y. Wada, T. Kastner, and M. J. Puma, "Groundwater depletion embedded in international food trade," *Nature*, vol. 543, pp. 700–704, 2017. DOI: 10.1038/nature21403.

[80] X. Kuang, J. Liu, B. R. Scanlon, *et al.*, "The changing nature of groundwater in the global water cycle," *Science*, vol. 383, no. 6686, eadf0630, 2024. DOI: 10.1126/science.adf0630.

[81] FAO, "Greenhouse gas emissions from agrifood systems – global, regional and country trends, 2000–2022," Food and Agriculture Organization of the United Nations, Rome, FAOSTAT Analytical Brief Series 94, 2024. [Online]. Available: https://openknowledge.fao.org/handle/20.500.14283/cd3167en.

[82] M. Crippa, E. Solazzo, D. Guizzardi, F. Monforti-Ferrario, F. Tubiello, and A. Leip, "Food systems are responsible for a third of global anthropogenic GHG emissions," *Nature Food*, vol. 2, no. 3, pp. 198–209, 2021. DOI: 10.1038/s43016-021-00225-9.

[83] X. Xu, P. Sharma, S. Shu, *et al.*, "Global greenhouse gas emissions from animal-based foods are twice those of plant-based foods," *Nature Food*, vol. 2, pp. 724–732, 2021. DOI: 10.1038/s43016-021-00358-x.

[84] J. Poore and T. Nemecek, "Reducing food's environmental impacts through producers and consumers," *Science*, vol. 360, no. 6392, pp. 987–992, 2018. DOI: 10.1126/science.aaq0216.

[85] M. A. Clark, N. G. G. Domingo, K. Colgan, *et al.*, "Global food system emissions could preclude achieving the 1.5° and 2°C climate change targets," *Science*, vol. 370, pp. 705–708, 2020. DOI: 10.1126/science.aba7357.

[86] A. Flammini, H. Adzmir, R. Pattison, K. Karl, Y. Allouche, and F. N. Tubiello, "Greenhouse gas emissions from cold chains in agrifood systems," *Sustainability*, vol. 16, no. 21, p. 9184, 2024. DOI: 10.3390/su16219184.

[87] United States Environmental Protection Agency, "Quantifying methane emissions from landfilled food waste," US EPA, EPA-600-R-23-064, 2023. [Online]. Available: https://www.epa.gov/land-research/quantifying-methane-emissions-landfilled-food-waste.

[88] J. Clapp, "Concentration and crises: Exploring the deep roots of vulnerability in the global industrial food system," *Journal of Peasant Studies*, vol. 50, no. 1, pp. 1–25, 2023. DOI: 10.1080/03066150.2022.2129013.

[89] USDA Foreign Agricultural Service, "Grain: World markets and trade," United States Department of Agriculture, Tech. Rep., Aug. 2025. [Online]. Available: https://apps.fas.usda.gov/psdonline/circulars/grain.pdf.

[90] J. Clapp, "The problem with growing corporate concentration and power in the global food system," *Nature Food*, vol. 2, no. 6, pp. 404–408, 2021. DOI: 10.1038/s43016-021-00297-7.

[91] FAO, "The state of food and agriculture 2025 — addressing land degradation across landholding scales," Food and Agriculture Organization of the United Nations, Rome, Tech. Rep., 2025. DOI: 10.4060/cd7067en.

[92] C. A. Monteiro, M. L. C. Louzada, E. Steele-Martinez, G. Cannon, G. C. Andrade, P. Baker, *et al.*, "Ultra-processed foods and human health: The main thesis and the evidence," *The Lancet*, vol. 406, pp. 2667–2684, 2025. DOI: 10.1016/S0140-6736(25)01565-X.

[93] C. A. Monteiro, J.-C. Moubarac, G. Cannon, S. W. Ng, and B. Popkin, "Ultra-processed products are becoming dominant in the global food system," *Obesity Reviews*, vol. 14, no. S2, pp. 21–28, 2013. DOI: 10.1111/obr.12107.

[94] M. M. Lane, E. Gamage, S. Du, *et al.*, "Ultra-processed food exposure and adverse health outcomes: Umbrella review of epidemiological meta-analyses," *BMJ*, vol. 384, e077310, 2024. DOI: 10.1136/bmj-2023-077310.

[95] M.-S. Fan, F.-J. Zhao, S. J. Fairweather-Tait, P. R. Poulton, S. J. Dunham, and S. P. McGrath, "Evidence of decreasing mineral density in wheat grain over the last 160 years," *Journal of Trace Elements in Medicine and Biology*, vol. 22, no. 4, pp. 315–324, 2008. DOI: 10.1016/j.jtemb.2008.07.002.

[96] D. R. Davis, M. D. Epp, and H. D. Riordan, "Changes in USDA food composition data for 43 garden crops, 1950 to 1999," *Journal of the American College of Nutrition*, vol. 23, no. 6, pp. 669–682, 2004. DOI: 10.1080/07315724.2004.10719409.

[97] R. J. Marles, "Mineral nutrient composition of vegetables, fruits and grains: The context of reports of apparent historical declines," *Journal of Food Composition and Analysis*, vol. 56, pp. 93–103, 2017. DOI: 10.1016/j.jfca.2016.11.012.

[98] A.-M. B. Mayer, L. Trenchard, and F. Rayns, "Historical changes in the mineral content of fruit and vegetables in the UK from 1940 to 2019: A concern for human nutrition and agriculture," *International Journal of Food Sciences and Nutrition*, vol. 73, no. 3, pp. 315–326, 2022. DOI: 10.1080/09637486.2021.1981831.

[99] R. M. Welch and R. D. Graham, "Breeding for micronutrients in staple food crops from a human nutrition perspective," *Journal of Experimental Botany*, vol. 55, no. 396, pp. 353–364, 2004. DOI: 10.1093/jxb/erh064.

[100] C. K. Khoury, A. D. Bjorkman, H. Dempewolf, *et al.*, "Increasing homogeneity in global food supplies and the implications for food security," *Proceedings of the National Academy of Sciences*, vol. 111, no. 11, pp. 4001–4006, 2014. DOI: 10.1073/pnas.1313490111.

[101] M. Springmann *et al.*, "Options for keeping the food system within environmental limits," *Nature*, vol. 562, pp. 519–525, 2018. DOI: 10.1038/s41586-018-0594-0.